

■ SDN-Based Honeypot Threat Detection & Secure Routing for Financial Networks

Intelligent Intrusion Detection, Real-Time DDoS Mitigation & SDN Rerouting

Domain: Financial Network Security & SDN

ML Engine: Isolation Forest & ExtraTree

Frameworks: Flask, Ryu/OpenFlow, Mininet

Architecture: 4-Tier Software-Defined Network

■■ System Architecture: 4-Tier SDN Model

Layer / Tier	Components	Role & Functionality
1. Network Edge	Customer (10.0.0.1) Attacker (192.168.100.x)	Legitimate financial traffic & simulated DDoS botnet sources.
2. Data Plane	OpenFlow Switches (s1, s2, s3, s4)	High-speed packet forwarding and dynamic flow-mod rule enforcement.
3. Control Plane	SDN Ryu Controller (flow_manager.py)	Centralized topology visibility, reactive flow routing, and path computation.
4. Application & ML	ML Anomaly Detector Flask SPA Dashboard Honeypot Trap (10.0.0.5)	Unsupervised threat classification, real-time live telemetry, and attacker entrapment.

■ Dual-Layer Threat Detection Engine

1. Machine Learning Anomaly Detection (Isolation Forest):

- Unsupervised learning trained on network flow statistics (packet rate, byte length, inter-arrival time).
- Automatically flags unknown volumetric DDoS floods (HTTP Flood, SYN Flood, UDP Flood) with anomaly score -1.

2. Deterministic Rule Matching:

- Fast pre-filter for known IP blacklists, signature patterns, and unauthorized port access attempts.
- Provides sub-millisecond classification for high-priority banking infrastructure protection.



Dynamic Secure Rerouting & Honeypot Decoy

Operational State	Visual Indicator	Network Flow Behavior
State 1: Normal	■ Emerald Stream	Customer (10.0.0.1) → Switches (s1/s2) → Bank Server (10.0.0.2). Direct low-latency path.
State 2: Attack Detected	■ Red Threat Alert	Attacker botnet flood identified by ML Anomaly engine. Bank link isolated.
State 3: Secure Rerouting	■ Purple Decoy Path	SDN controller pushes OpenFlow Flow-Mod rules: Attacker diverted to Honeypot (10.0.0.5:8000), Customer traffic remains 100% operational.

■ Verification & Demonstration Results

Key Demonstration Achievements:

- **Zero Downtime for Banking API:** Legitimate transactions processed continuously during active 14-burst botnet DDoS attacks.
- **Automated Isolation:** Attacker IP addresses (192.168.100.101–114) automatically trapped in decoy honeypot VLAN without human intervention.
- **Real-Time Live UI Synchronization:** Single Page Application dashboard updates stats, SVG animated particle paths, threat tables, and honeypot forensics in under 1 second.
- **Comprehensive Forensics:** All attack signatures, request payloads, and flow modifications logged and exportable to CSV.

Project: SDN-Based Honeypot Threat Detection & Secure Routing for Financial Networks